COURSE: BAMAT205 | MODULE 3: CODING THEORY | LINEAR ALGEBRA & DISCRETE MATHEMATICS

BitShield: Hamming (7,4) Error Detection & Correction

Complete Mathematical Specification, Algorithmic Implementation, & Viva Voce Dossier

ACADEMIC PROJECT	REPOSITORY	LIVE APP	VERIFICATION
BitShield System	gamerboyadarsh-dot/Bit-Shield	gamerboyadarsh-dot.github.io	173 / 173 Vitest Suite Passed

01 Problem Statement & Engineering Purpose

In digital communications and physical storage devices (dynamic RAM, flash memory, deep-space telemetry), binary data is continuously subject to physical disturbances such as cosmic ray ionization, electrical thermal noise, and electromagnetic interference. These physical anomalies invert bits from **0** → **1** or **1** → **0**.

Without redundancy, a receiver has no way of verifying data integrity. The objective of **Forward Error Correction (FEC)** is to encode a k-bit message by appending $r = n - k$ parity check bits. This forms an n-bit codeword that allows the receiver to detect and rectify errors autonomously without requiring a retransmission.

The Coding Theory Objective: Hamming (7,4)

Encode $k = 4$ information bits into an $n = 7$ bit codeword using $r = 3$ parity bits. The code guarantees 100% detection and correction of any single-bit error ($t = 1$), and detects all double-bit errors under SECDED (8,4) extension.

02 Vector Space Foundations over Galois Field GF(2)

All operations are strictly executed over the finite field $\text{GF}(2) = (\{0, 1\}, \oplus, \cdot)$, where addition is arithmetic modulo 2 (bitwise **XOR**) and multiplication is arithmetic modulo 2 (bitwise **AND**).

GALOIS FIELD ARITHMETIC AXIOMS

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0 ⊕ 0 = 0  |  0 ⊕ 1 = 1  |  1 ⊕ 0 = 1  |  1 ⊕ 1 = 0  (Self-inverse: -x = x)
0 · 0 = 0  |  0 · 1 = 0  |  1 · 0 = 0  |  1 · 1 = 1
```

The vector space of all possible 7-bit binary words is $V = \text{GF}(2)^7$, containing $|V| = 2^7 = 128$ vectors. The Hamming (7,4) code C constitutes a **4-dimensional linear subspace** of $\text{GF}(2)^7$, containing exactly $|C| = 2^4 = 16$ valid codewords.

03 Mathematical Conventions & Bit Allocation

Bit positions are indexed 1 through 7 from left to right. Codeword vector:

$$c = (c_1, c_2, c_3, c_4, c_5, c_6, c_7) = (p_1, p_2, d_3, p_4, d_5, d_6, d_7)$$

- **Parity bits (p_1, p_2, p_4):** Placed at indices that are powers of 2 (positions 1, 2, and 4), corresponding to single-bit binary masks 001_2, 010_2, and 100_2.
- **Data bits (d_3, d_5, d_6, d_7):** The 4-bit message $m = (m_1, m_2, m_3, m_4)$ maps directly to indices 3, 5, 6, and 7.

EVEN PARITY ENCODING FORMULAS

$p_1 = d_3 \oplus d_5 \oplus d_7 = m_1 \oplus m_2 \oplus m_4$ (Covers positions with bit 0 set: 1, 3, 5, 7)
$p_2 = d_3 \oplus d_6 \oplus d_7 = m_1 \oplus m_3 \oplus m_4$ (Covers positions with bit 1 set: 2, 3, 6, 7)
$p_4 = d_5 \oplus d_6 \oplus d_7 = m_2 \oplus m_3 \oplus m_4$ (Covers positions with bit 2 set: 4, 5, 6, 7)

04 Matrix Formulation: Generator & Parity-Check Matrices

Linear block codes are completely described by two fundamental matrices over $\text{GF}(2)$: the Generator Matrix G and Parity-Check Matrix H.

GENERATOR MATRIX G (4 × 7) : C = M · G

	p_1	p_2	d_3	p_4	d_5	d_6	d_7
m_1	1	1	1	0	0	0	0
m_2	1	0	0	1	1	0	0
m_3	0	1	0	1	0	1	0
m_4	1	1	0	1	0	0	1

PARITY-CHECK MATRIX H (3 × 7) : H · C^T = 0

	1	2	3	4	5	6	7
S_4	0	0	0	1	1	1	1
S_2	0	1	1	0	0	1	1
S_1	1	0	1	0	1	0	1

The Fundamental Linear Algebra Beauty of H

Column j of matrix H (for $j = 1, 2, \dots, 7$) is the exact 3-bit binary representation of decimal index j, read vertically as $[S_4, S_2, S_1]^T$. Furthermore, $G \cdot H^T = \mathbf{0}_{4 \times 3} \pmod 2$, proving the code space C is the null space of H.

05 Channel Transmission, Syndrome Decoding & Correction

Let $c \in C$ be the transmitted codeword. Physical noise acts additively: $r = c \oplus e$, where $e \in \text{GF}(2)^7$ is the error vector.

> **SYNDROME EVALUATION PROOF**
>
> $$s = H \cdot r^T = H \cdot (c \oplus e)^T = (H \cdot c^T) \oplus (H \cdot e^T) = 0 \oplus H \cdot e^T = H \cdot e^T$$

Notice that the syndrome s depends **exclusively on the channel noise e** and is completely invariant to the message content c.

> **RECEIVER SYNDROME SCALAR EQUATIONS**
>
> $$S_1 = r_1 \oplus r_3 \oplus r_5 \oplus r_7 \quad \text{(Parity Check on Group 1)}$$
> $$S_2 = r_2 \oplus r_3 \oplus r_6 \oplus r_7 \quad \text{(Parity Check on Group 2)}$$
> $$S_4 = r_4 \oplus r_5 \oplus r_6 \oplus r_7 \quad \text{(Parity Check on Group 4)}$$
>
> $$\text{Decimal Error Position} = 4 \cdot S_4 + 2 \cdot S_2 + 1 \cdot S_1$$

Correction Rule: If the decimal syndrome equals 0, no single error is detected. Otherwise, the syndrome reveals the exact index j. The corrected vector is:

$$\hat{c} = r \oplus e_j \quad \textbf{(Invert bit at index j)}$$

06 Rigorous Theoretical Proofs

Theorem 1: Minimum Hamming Distance d_min = 3

Proof: The minimum distance of a linear block code equals the minimum weight of its non-zero codewords, which equals the minimum number of linearly dependent columns in H.
1. All columns of H are non-zero $\implies d_{\min} > 1$.
2. All columns of H are pairwise distinct. Over $\text{GF}(2)$, two vectors are linearly dependent if and only if they are identical. Since no two columns are equal, every pair of columns is linearly independent $\implies d_{\min} > 2$.
3. Columns 1, 2, and 3 satisfy: $\text{Col}_1 \oplus \text{Col}_2 = [0, 0, 1]^T \oplus [0, 1, 0]^T = [0, 1, 1]^T = \text{Col}_3$. This demonstrates a set of 3 linearly dependent columns. Hence, $d_{\min} = 3$. $\blacksquare$

Theorem 2: Hamming (7,4) is a "Perfect Code"

Proof: The Sphere-Packing Bound for a code correcting $t = 1$ errors is:

> $$2^k \cdot [\ C(n, 0) + C(n, 1)\ ] \leq 2^n$$
> $$2^4 \cdot [\ 1 + 7\ ] = 16 \cdot 8 = 128 = 2^7$$

The inequality holds with **exact equality**. The 16 Hamming spheres of radius $t = 1$ centered around the 16 valid codewords completely and partition the entire 128-vector space with zero overlapping or empty elements. $\blacksquare$

Theorem 3: Coset Decomposition (Lagrange's Theorem)

The additive group $(\text{GF}(2)^7, \oplus)$ of order 128 is partitioned by the linear code subgroup C of order 16 into exactly $2^{7-4} = 8$ mutually disjoint cosets of size 16:

$$V = C \cup (C \oplus e_1) \cup (C \oplus e_2) \cup \ldots \cup (C \oplus e_7)$$

Each coset maps bijectively to one of the 8 unique 3-bit syndrome vectors. The coset leaders correspond to the minimum-weight error vectors $\mathbf{0}$ and $e_1, \dots, e_7$.

07 Multi-Bit Error Regimes & SECDED (8,4) Extension

Error Regime	Physical Condition	Standard (7,4) Response	SECDED (8,4) Response
0 Bit Errors	$e = \mathbf{0}$	Syndrome = 000: Correctly says "No error"	$s = 000, P_{\text{overall}} = 0$: Valid data
1 Bit Error	$e = e_j, \text{wt}(e) = 1$	Syndrome = $\text{bin}(j)$: **Corrected**	$s \neq 000, P_{\text{overall}} = 1$: **Corrected**
2 Bit Errors	$\text{wt}(e) = 2$	Syndrome points to 3rd clean bit: **Miscorrection!**	$s \neq 000, P_{\text{overall}} = 0$: **Double Error Alert**
3 Bit Errors	$\text{wt}(e) = 3$ (valid c')	Syndrome = 000: **Undetected Error**	$s = 000, P_{\text{overall}} = 1$: Parity fault detected

08 The Complete 16-Codeword Codebook

Dec	Message (m_1 m_2 m_3 m_4)	Parity (p_1 p_2 p_4)	Codeword (p_1 p_2 d_3 p_4 d_5 d_6 d_7)	Hamming Weight
0	`0000`	`000`	`0000000`	0
1	`0001`	`111`	`1101001`	4
2	`0010`	`011`	`0101010`	3
3	`0011`	`100`	`1000011`	3
4	`0100`	`101`	`1001100`	3
5	`0101`	`010`	`0100101`	3
6	`0110`	`110`	`1100110`	4
7	`0111`	`001`	`0001111`	4
8	`1000`	`110`	`1110000`	3
9	`1001`	`001`	`0011001`	3
10	`1010`	`101`	`1011010`	4
11	`1011`	`010`	`0110011`	4
12	`1100`	`011`	`0111100`	4
13	`1101`	`100`	`1010101`	4
14	`1110`	`000`	`0010110`	3
15	`1111`	`111`	`1111111`	7

09 Software Architecture & Technical Implementation

- **Mathematical Core (`src/lib/hamming.js`):** Completely pure functional module with 0 DOM and 0 React dependencies. Exports **`encode`**, **`syndrome`**, **`correct`**, **`flipBit`**, **`classify`**, **`evaluateSecded`**, **`generateCosets`**, **`simulateBscNoise`**.
- **Unit Testing Suite (`src/lib/hamming.test.js`):** 173 test assertions across 10 suites executed in 16ms with Vitest, guaranteeing 100% mathematical correctness against reference tables.
- **Accessibility & Dual-Encoding:** Parity bits use teal and diamond badges; data bits use blue and square badges; corrupted bits use red and alert symbols; corrected bits use amber and checkmarks. No color-blindness reliance.
- **Single-File Bundling:** Built using **`vite-plugin-singlefile`** producing a ~892 kB offline executable (**`dist/index.html`**) that runs without internet access.

10 Viva Voce Examination Cheat Sheet

Q1: Why are the parity bits assigned to positions 1, 2, and 4 instead of positions 1, 2, and 3?

Answer: Positions 1, 2, and 4 correspond to $2^0, 2^1, 2^2$ (binary **`001`**, **`010`**, **`100`**). Each contains exactly one binary set bit. Consequently, the binary expansion of every index $1..7$ corresponds to a unique subset of parity bits, allowing the syndrome (S_4, S_2, S_1) to directly form the binary address of the corrupted bit.

Q2: What is the physical meaning of $H cdot c^T = mathbf{0}$?

Answer: It establishes that the code space C is the null space (kernel) of linear transformation $H: ext{GF}(2)^7 o ext{GF}(2)^3$. A vector c is a valid codeword if and only if it satisfies all three simultaneous parity equations.

Q3: Why does standard Hamming (7,4) fail on double-bit errors?

Answer: Because $d_{min} = 3$. A double-bit error produces a syndrome that equals the sum of two columns of H. Because H contains all non-zero 3-bit vectors, that sum equals a third column of H. The decoder misinterprets this as a single error at that third location, flipping an uncorrupted bit and resulting in a 3-bit error (miscorrection).

Q4: How does SECDED resolve double-bit errors?

Answer: SECDED adds an 8th parity bit $p_8 = □igoplus_{i=1}^7 c_i$, elevating d_{min} to 4. When two bits flip, the 3-bit syndrome is non-zero, but the overall parity check succeeds ($P_{ ext{overall}} = 0$). This combination uniquely signals an uncorrectable double error and triggers a retransmission alert.